9. $\begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 1 & -3 & -1 \\ 0 & 0 & 0 & 2 & 4 \end{bmatrix}$ Row3 = Row3 x (2)/3 $\rightarrow \begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 2/3 & -2 & -2/3 \\ 0 & 0 & 0 & 2 & 4 \end{bmatrix}$

Row3 = Row3 + Row4 $\begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 2/3 & 0 & 10/3 \\ 0 & 0 & 0 & 2 & 4 \end{bmatrix}$

Row4 = Row4 x $\frac{1}{2}$ $\begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 2/3 & 0 & 10/3 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$

Row3 = Row3 x $\frac{3}{2}$

$\begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & -3 & 0 & -7 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$ Row2 = Row2 + Row3 x 3 $\begin{bmatrix} 1 & -1 & 0 & 0 & -4 \\ 0 & 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$

Row1 = Row1 + Row2 $\begin{bmatrix} 1 & 0 & 0 & 0 & 4 \\ 0 & 1 & 0 & 0 & 8 \\ 0 & 0 & 1 & 0 & 5 \\ 0 & 0 & 0 & 1 & 2 \end{bmatrix}$ $\boxed{x_1 = 4 \quad x_2 = 8 \quad x_3 = 5 \quad x_4 = 2}$

Solve the systems in Exercises 11-14

11) $x_2 + 4x_3 = -5$
$x_1 + 3x_2 + 5x_3 = -2$
$3x_1 + 7x_2 + 7x_3 = 6$

Augmented matrix: $\begin{bmatrix} 0 & 1 & 4 & -5 \\ 1 & 3 & 5 & -2 \\ 3 & 7 & 7 & 6 \end{bmatrix}$

Interchange row 3 and row 1 $\rightarrow \begin{bmatrix} 3 & 7 & 7 & 6 \\ 1 & 3 & 5 & -2 \\ 0 & 1 & 4 & -5 \end{bmatrix}$

Row1 = Row1 - 3x row 2 $\begin{bmatrix} 0 & -2 & -8 & 12 \\ 1 & 3 & 5 & -2 \\ 0 & 1 & 4 & -5 \end{bmatrix}$ Interchange row 1 and row 2 $\rightarrow \begin{bmatrix} 1 & 3 & 5 & -2 \\ 0 & -2 & -8 & 12 \\ 0 & 1 & 4 & -5 \end{bmatrix}$

Row2 = Row2 x $\frac{1}{2}$